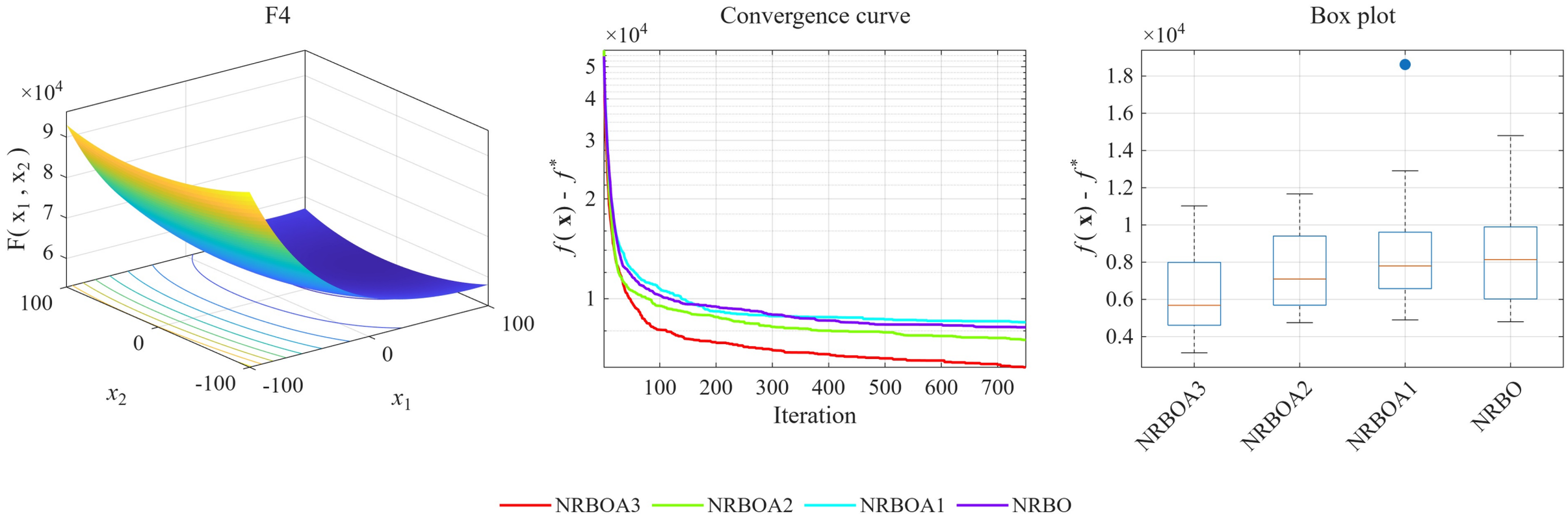

F4
×10^4
F(x1 , x2)
9
8
7
6
100
0
-100
x2
-100
0
100
x1
Convergence curve
×10^4
5
4
3
2
1
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO